

Janit Rajkarnikar

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EDUCATION

University of Southern Mississippi

Bachelor of Science in Computer Science, Minor in Mathematics and Data Analysis

GPA: 4.0

May 2026

Hattiesburg, MS

WORK AND RESEARCH EXPERIENCE

Software Engineer Intern

Aug 2024 – Present

Woafmeow, Inc

Remote

- Orchestrated a comprehensive backend restructure, implementing a modular layered architecture with FastAPI and SQLAlchemy ORM that improved code maintainability, leveraging PostgreSQL for robust data management and advanced ORM techniques
- Designed and implemented an innovative daily tasks and points system, creating a gamification mechanism that increased user engagement by 30% and drove consistent platform interaction through strategic UI/UX design and interactive reward mechanisms
- Developed and implemented robust security infrastructure utilizing JWT authentication, creating a multi-layered authentication system with role-based access control, token generation, and refresh mechanisms to ensure comprehensive endpoint protection, user data privacy, and secure application access

Research Assistant

May 2024 – Dec 2024

University of Southern Mississippi

Hattiesburg, MS

“Ensemble Machine Learning Approach to Phishing Website Detection,” approved for the EquinOCS system.

- Developed ensemble machine learning models for phishing website detection, leveraging Bagging, Boosting, and Stacking to achieve 98.66% detection accuracy with minimized false positives.
- Created a feature selection pipeline utilizing Scikit-learn’s SelectKBest and ANOVA F-tests, reducing dataset features from 112 to 40, enhancing both model scalability and real-time applicability.
- Conducted comprehensive performance benchmarking across multiple ML models, implementing hyperparameter optimization via GridSearchCV to maximize precision, recall, and F1 scores.

PROJECTS

The Designer’s Touch

React, Node.js, Express, Three.js, Firebase, AWS, Azure

- Developed a web platform using React, Node.js, and Three.js, enabling users to create personalized merchandise with interactive, real-time 3D visualization.
- Deployed the platform using AWS S3 and Azure, implementing continuous integration and delivery pipelines while maintaining server reliability.
- Built a secure and user-friendly interface with React, Firebase, and Express, allowing users to log in, create and customize artwork (designs), apply it to different sections of apparel, save their creations, leverage AI-generated design suggestions, and complete purchases through a robust checkout system.

Emotion Tracker (Emoki)

React Native, Node.js, Flask, MongoDB, Firebase

- Implemented real-time emotion detection using RoBERTa models, providing insights into emotional trends and logging intense emotions along with location data for detailed analysis of triggers.
- Designed a chatbot feature that analyzes user messages, tracks emotions, and maintains a personalized diary to help users reflect and manage emotional well-being.
- Optimized the backend with Flask and Express for seamless data processing and integrated Firebase for secure authentication and MongoDB for real-time data synchronization.

AFM Image Processing Application

C#, .NET, WPF, Python

- Developed a desktop application using .NET 8.0 (WPF) and pythonnet to seamlessly integrate Python-based image processing of AFM .ibw files—no external server required.
- Created robust Python scripts to fully automate AFM .ibw file processing, incorporating data extraction via Igor, advanced statistical modeling with scikit-learn for plane fitting, polynomial leveling, and masking large features using NumPy arrays.
- Leveraged matplotlib for high-quality image generation and grid layouts, ensuring a fully integrated, reproducible, and efficient pipeline for transforming raw microscopy data into ready-to-use, visually informative results.

TECHNICAL SKILLS

Programming Languages: JavaScript, C++, Python, C#, Visual Basic

Libraries/Frameworks: Node.js, Django, React, Vue.js, Next.js, Ruby on Rails, Three.js, React Native, Flask, PyTorch, .NET, Matplotlib, pandas, Unity

Tools/Platforms: Git, Firebase, AWS, Azure, Expo, WPF

Databases: MySQL, SQLite, MongoDB, PostgreSQL